

The Quantum Ecosystem of the Future

It will still take about a decade for quantum computing technology to be a part of our daily lives. Nevertheless, one of the most active areas of research and development nowadays is quantum computing. That's because the adoption of quantum processing units (QPUs) for both application-specific and general-purpose computing is likely as they develop from being just simple scientific testbeds to reliable devices.

The term 'quantum chip,' or 'quantum processing unit,' or 'QPU' refers to a physical (fabricated) chip that has several interconnected qubits, or quantum bits. A QPU serves as the basic building block of a complete quantum computer, which also consists of the control electronics, the QPU's housing environment, and several other parts. A QPU enables a form of computation that's built on quantum physics—nature's operating system.

How a QPU differs from a CPU

A quantum computer (with QPU) operates in a fundamentally different way from a classical computer. A

'bit' in traditional computing (using a CPU) is a very small unit of binary data, which can be either a one or a zero. A processed bit is implemented by one of two levels of low DC voltage in traditional computer systems. The most recent variation of this concept is the 'qubit,' or quantum bit. Qubits use the phenomenon of superposition, which enables them to be in numerous states simultaneously, in contrast to binary bits, which can only be in either/or situations.

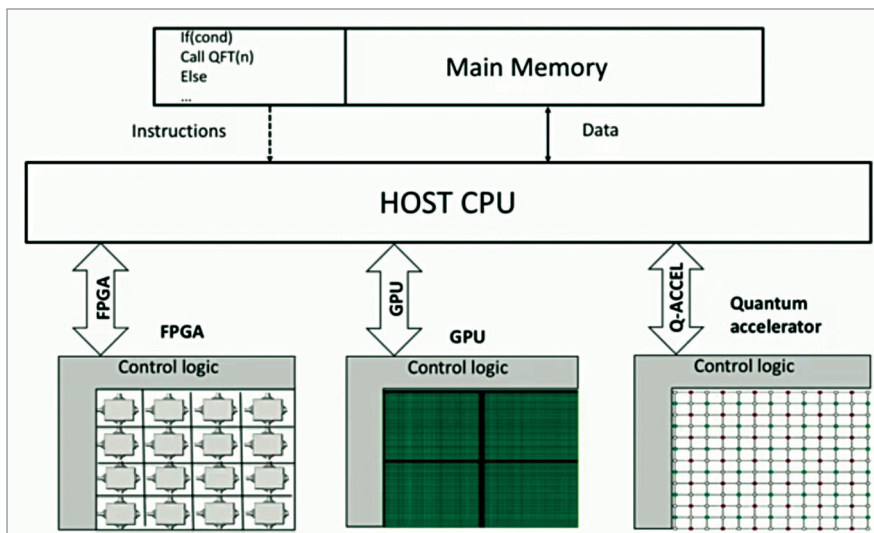
The capabilities of quantum computers surpass those of traditional computers, especially in terms of speed. A quantum computer can complete a task in two or three

stages, whereas a classical computer needs thousands of steps to handle the same problem. Therefore, most quantum chips or QPUs are used as accelerators in heterogeneous multi-core computing devices. These can be connected with a classical processor (CPU) to provide a performance boost that cannot be achieved classically.

The classical control circuitry and the quantum processing unit (QPU) are two of the most important components of hardware in quantum computing. Some of the subsystems of the QPU include registers and gates (sometimes called QRAM), a quantum control unit for driving the system states, and circuitry to interface between the classical host CPU and the QPU. However, there are various types of QPUs with variations in the subsystems as well as the underlying principles.

Types of QPU hardware

Spin based: The spin qubit quantum computer works by manipulating the spin of electrons and electron holes, which are the charge carriers in semiconductor devices. Here, individual charges and their associated spins are trapped in semiconductor quantum dots (QDs), which are subsequently utilised as qubits. Leti, Intel, Imec and others are working on this technology.



An example of a heterogeneous multi-core system with a quantum co-processor that provides a substantial increase in computational power (Credit: QuTech Academy)